



FITNESSWISE

Ramji Srinivasan



The precarious yet rewarding world of plyometrics

Plyometric training protocols have **tremendous benefits.** However, there are common mistakes that need to be avoided while administering them.

The term 'plyometrics' comes from the Greek words *pleion* (meaning more) and *metron* (meaning measure). Plyometrics is a training method that focuses on the stretch reflex of muscles through jumping, bounding, and other explosive movements. It teaches athletes how to exert the maximum amount of force in a short amount of time and is an excellent way to build explosive power.

It's true definition is the rapid stretching of a muscle (eccentric action) immediately followed by a powerful and fast shortening of the same muscle and connective tissue (concentric action). T

his shortening uses stored elastic energy (also known as the Stretch Shortening Cycle) to produce greater force compared to a concentric action alone.

Research shows that adding plyometrics to a strength training programme can improve vertical jump, acceleration, muscular power, leg strength, and overall proprioception.

But the important question is: How to do it, and when to incorporate it?

There are tremendous benefits to be gained from correctly administering plyometric training protocols. When designing and implementing a plyometric programme to train athletes for greater explosiveness, several factors must be considered. A solid understanding of the science behind this type of training is essential to maximise benefits.

Too often, trainers prescribe exercises without fully understanding their purpose, which can result in more harm than good when using plyometric training.

While the exercises themselves might be effective, using incorrect sets, reps, inadequate rest, or improper

tempo can produce sub-par results. In such cases, results are well below expectations for both the athlete and the trainer. Additionally, poor programming can lead to injuries and prevent athletes from reaching peak performance at the right time.

Many athletes avoid plyometrics due to fear of injury. However, plyometric training is an important component of injury prevention programs.

It's essential to train the body to adapt to the jumping and landing demands of a sport before the season begins.

It's best to start by incorporating low-intensity plyometrics into warm-ups, then progress to a low volume of medium-intensity exercises. The progression should be tailored to the sport and the athlete's skill level.

The main goal of the program should be to apply the optimal amount of stress, provide sufficient recovery time between exercises and sessions, and design a plan that amplifies the training effect.

Let's discuss several common mistakes trainers or coaches make in the design and execution of their plyometric training programs:

1. Lack of understanding of key